



CLEMENTI TOWN SECONDARY SCHOOL
SECONDARY 1: G3 MATHEMATICS
WA 3 REVISION Set 2

Name: _____ ()

Class: _____

Chapter 4: Basic Algebra and Algebraic Manipulations

Chapter 5: Simple Equations in One Variable

Chapter 10: Sequences & Number Patterns

Chapter 11: Linear Functions and Graphs

1 Write down the next two terms of each sequence.

(a) $-30, -36, -42, -48, \dots$

[1]

(b) $\frac{2}{3}, \frac{5}{8}, \frac{13}{21}, \frac{34}{55}, \dots$

[1]

- 2 Consider the first four expressions of the following pattern:

$$T_1 = 1^3 - 5$$

$$T_2 = 2^3 - 9$$

$$T_3 = 3^3 - 13$$

$$T_4 = 4^3 - 17$$

$\vdots$

- (a) Write down and evaluate the value of T_8 . [2]

- (b) Write down an expression, in terms of n , for T_n . [1]

- (c) Timothy says that the second term in the expression of T_n is always odd.
Do you agree? Explain your answer. [1]

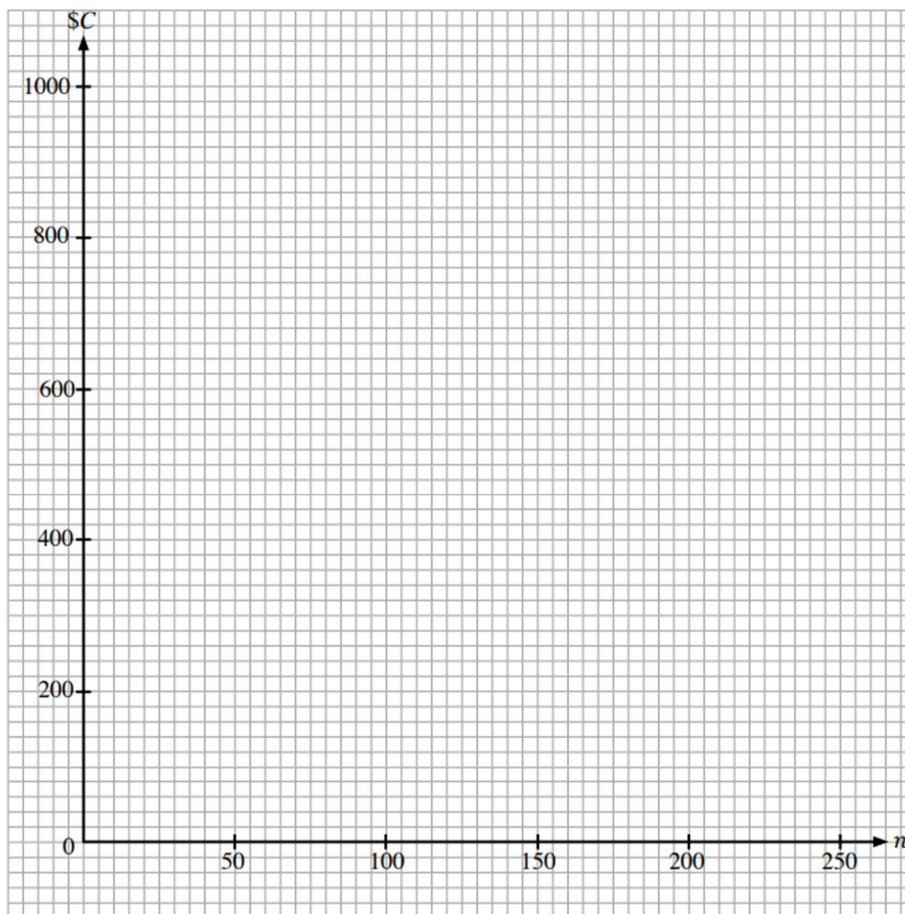
- 3 The cost, $\$C$, of printing n books is given by the function $C = 3n + 200$.

(a) Complete the table below for some values of C and n .

[1]

n	100	150	200
C			

(b) On the given grid below, plot the graph of $C = 3n + 200$ for values of n between 0 and 200 inclusive. [2]



(c) Find the cost per book when printing 180 books.

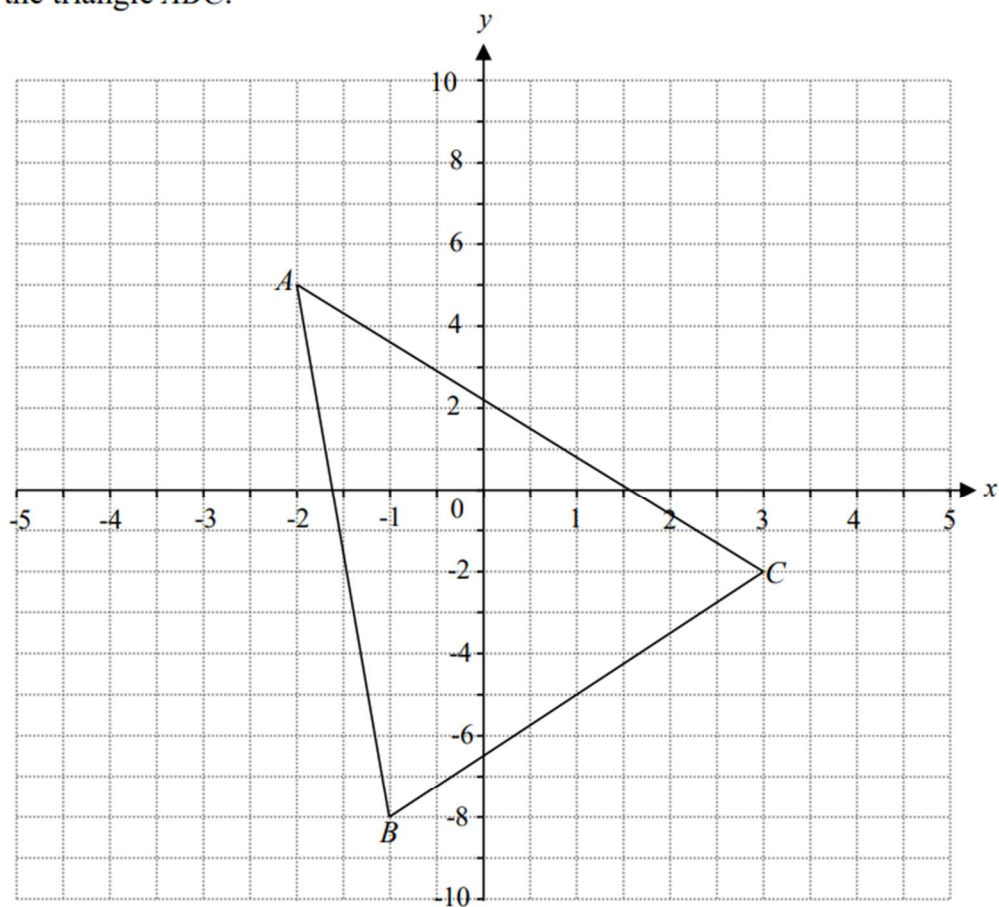
[1]

(d) What does the constant term 200 represent?

[1]

4

The points A , B and C on the Cartesian plane below are connected by straight lines to form the triangle ABC .



- (a) Which line has a positive gradient? Explain your answer.
- (b) Hence, find the gradients of this line.
- (c) Find the area of triangle ABC .

5 Solve the following equations.

(a) $2(5x - 8) - 7 = -3(-5 - 4x)$

[2]

(b) $\frac{-3x}{x-2} = \frac{8}{3}$

[2]

(c) $\frac{2x-5}{4} - \frac{4-x}{3} + \frac{x}{6} = 0$

[3]

- 6 David has some money to buy fruits.
He can buy n papayas at \$3.80 each and have \$3.40 left.
He can otherwise buy $(n + 18)$ apples at \$0.70 each and have \$0.10 left.
Find the amount of money David has.

[3]

- 7 Express the following as a single fraction in its simplest form.

[2]

$$\frac{1+x}{3} - \frac{2(x-3x)}{4} - y$$

- 8 Factorise the following expressions completely.

(a) $12xy - 4yz$

[1]

(b) $28c + 20ac - 26bc^2$

[1]

- 9 **(a)** Write down an algebraic expression for the following statement.
Twice the sum of $6a$ and cube of x divided by the square of mn . [1]

- (b)** Find the fraction that is exactly halfway between $\frac{23}{47}$ and $\frac{24}{47}$. [1]

- 10 Simplify $6(3m - n) - 2(5n - 3m + x)$. [2]

ANSWER – Set 2

1(a)	-54, -60 Subtract 6 from each term to obtain the next								
1(b)	$\frac{89}{144}, \frac{233}{377}$ Add the previous numerator / denominator to the current denominator / numerator								
2(a)	$T_8 = 8^3 - 33$								
2(b)	$T_n = n^3 - (4n + 1)$ or $T_n = n^3 - 4n - 1$								
2(c)	Yes agree. 4 is even, $4n$ is always even, $4n + 1$ will thus be odd.								
3(a)	<table><tr><td>n</td><td>100</td><td>150</td><td>200</td></tr><tr><td>C</td><td>500</td><td>650</td><td>800</td></tr></table>	n	100	150	200	C	500	650	800
n	100	150	200						
C	500	650	800						
3(b)									
3(c)	From the graph, $C = 740$ It will cost \$740.								
3(d)	It represents the fixed cost when no books are printed.								
4	(a) BC (b) $\frac{3}{2}$ (c) 29 units^2								
5(a)	$2(5x - 8) - 7 = -3(-5 - 4x)$								

	$10x - 16 - 7 = 15 + 12x$ $-2x = 38$ $x = -19$
5(b)	$\frac{-3x}{x-2} = \frac{8}{3}$ $-9x = 8x - 16$ $x = \frac{16}{17}$
5(c)	$\frac{2x-5}{4} - \frac{4-x}{3} + \frac{x}{6} = 0$ $\frac{3(2x-5) - 4(4-x) + 2x}{12} = 0$ $3(2x - 5) - 4(4 - x) + 2x = 0$ $6x - 15 - 16 + 4x + 2x = 0$ $12x = 31$ $x = 2\frac{7}{12}$
6	$3.8n + 3.4 = 0.7(n + 18) + 0.1$ $3.8n + 3.4 = 0.7n + 12.6 + 0.1$ $3.1n = 9.3$ $n = 3$ Amount of money David has $= 3.8(3) + 3.4$ $= \$14.80$
7	$\frac{1+x}{3} - \frac{2(x-3y)}{4} - y$ $= \frac{4(1+x) - 6(x-3y) - 12y}{12}$ $= \frac{4+4x-6x+18y-12y}{12}$ $= \frac{4-2x+6y}{12}$ $= \frac{2-x+3y}{6}$
8(a)	$12xy - 4yz = 4y(3x - z)$
8(b)	$28c + 20ac - 26bc^2 = 2c(14 + 10a - 13bc)$
9(a)	$\frac{2(6a+x^3)}{(mn)^2} \text{ or } \frac{12a+2x^3}{(mn)^2}$
9(b)	$\left(\frac{23}{47} + \frac{24}{47}\right) \div 2 = 0.5$
10	$6(3m - n) - 2(5n - 3m + x)$ $= 18m - 6n - 10n + 6m - 2x$ $= 24m - 16n - 2x$